University of Central Punjab

Information Security

Assignment 1

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Information Security - Assignment #1

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-: Question # 1 :-

: Rail Fence Cipher :

(a)

Plaintext = CYBERSECURITYMATTERS

No. of rails = 3

C			R			U			Y			T		
	Y		E	S		C	R		T	M		T	E	S
		B			E			I			A			R

Rail 1: CRUYT

Rail 2: YESCRTMTES

Rail 3: BEIAR

C = CRUYT + YESCRTMTES + BEIAR

Ciphertext = CRUYTYESCRTMTESBEIAR

(b)

Ciphertext = PTARCYDTOEORATU

No. of Rails = 4

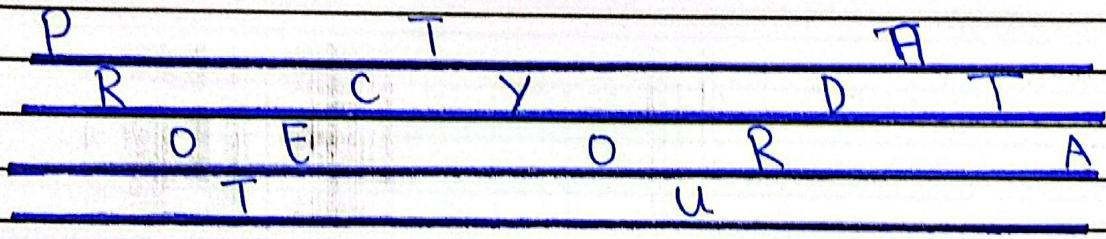
Total Characters = 15

The zigzag pattern for 4 rails follows:
Rail 1 → Rail 2 → Rail 3 → Rail 4 → Rail 3 → Rail 2 → Rail 1 → ...

For 15 characters, the number of positions on each rail is:

Rail	Position	Count
Rail 1	1, 7, 13	3
Rail 2	2, 6, 8, 12, 14	5
Rail 3	3, 5, 9, 11, 15	5
Rail 4	4, 10	2

Rail 1 : PTA
Rail 2 : RCYDT
Rail 3 : OEORA
Rail 4 : TU



P → R → O → T → E → C → T → Y → O → U → R → D → A → T → A

Plaintext = PROTECTYOURDATA

(c)

: Vulnerability of RF-C to Brute-Force Attacks :

Ans: The Rail Fence Cipher is vulnerable to brute-force attacks because it has a very small key space. The key is simply the number of rails. An attacker can try different rail counts until meaningful plaintext is obtained. Also, the cipher only rearranges characters without changing them, which makes it easier to break.

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:- Question No. 2 :-

: Columnar Transposition Cipher :

(a)

Plaintext = NETWORKSECURITYISIMPORTANT
Key = CLOUD

The key "CLOUD" contains 5 letters.
Total plaintext letters = 26.

$$\text{Rows required} = \left\lceil \frac{26}{5} \right\rceil = 6$$

Total spaces = 6×5 = 30
Padding Required = 30−26 = 4 characters.

Padded Plaintext = NETWORKSECURITYISIMPORTANTXXXX

C	L	O	U	D
N	E	T	W	O
R	K	S	E	C
U	R	I	T	Y
I	S	I	M	P
O	R	T	A	N
T	X	X	X	X

Column Numbering:

C	L	O	U	D
1	3	4	5	2

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Key	Column Letters
C (1)	NRUIOT
P (2)	OCYPNX
L (3)	EKRSRX
O (4)	TSIITX
U (5)	WETMAX

C	D	L	O	U
N	O	E	T	W
R	C	K	S	E
U	Y	R	I	T
I	P	S	I	M
O	N	R	T	A
T	X	X	X	X

Ciphertext = NRUIOT+OCYPNX+EKRSRX+TSIITX+WETMAX

→ NRUIOTOCYPNXEKRSRXTSIITXWETMAX

(b)

Ciphertext = ERCCRWASNL TKEOONOCST
Key = SAFE

Ciphextext length = 20
No. of Columns = 4
Rows = $\frac{20}{4} = 5$

Key alphabetically = A, E, F, S

S	A	F	E
4	1	3	2

Key in order	Ciphertext Group
A	ERCCR
E	WASNL
F	TKEOO
S	NOCST

S	A	F	E
N	E	T	W
O	R	K	A
C	C	E	S
S	C	O	N
T	R	O	L

Row 1 = NETW
Row 2 = ORKA
Row 3 = CCES
Row 4 = SCON
Row 5 = TROL

Plaintext = NETWORK ACCESS CONTROL

(c)

: Importance of Key & Column Ordering :

Ans: The key determines the order in which columns are read during encryption. During decryption, the same key is used to place the columns back in their original positions. If the wrong key or column order is used, the decrypted message will be incorrect. Therefore, the key is essential for correctly encrypting and decrypting the message.

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Question # 03

Frequency Analysis:

Cipher text:

OYR NRCPMAOX ORSG PNRN NOMJHT KSNNVJMUN OJ KMJORCO USOS. RQRMX PNRM GPNO MRKJMO NPNKACAJPN RGSAFN

(a)

Total letters = 131

Letter	Count	Frequency
N	19	19/131 x100 = 14.5%
R	18	18/131 x100 = 13.7%
O	13	13/131 x100 = 9.9%
M	12	12/131 x100 = 9.2%
J	10	10/131 x100 = 7.6%
P	7	7/131 x100 = 5.3%
S	7	7/131 x100 = 5.3%
C	6	6/131 x100 = 4.6%
A	6	6/131 x100 = 4.6%
K	5	5/131 x100 = 3.8%
X	4	4/131 x100 = 3.1%
V	4	4/131 x100 = 3.1%
G	3	3/131 x100 = 2.3%
H	3	3/131 x100 = 2.3%
F	3	3/131 x100 = 2.3%
Y	2	2/131 x100 = 1.5%
U	2	2/131 x100 = 1.5%

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Q	2	1.5%
I	2	1.5%
T	1	1/131 x 100 = 0.8%
D	1	1/131 x 100 = 0.8%
E	1	1/131 x 100 = 0.8%

* Five most frequent letters:

N (19, 14.5%)
R (18, 13.7%)
O (13, 9.9%)
M (12, 9.2%)
J (10, 7.6%)

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(b)

most common letters:

E, T, A, O, I, N

Comparing with common english frequencies.

Cipher letters	P.T	Confirmation
N	S	The word is "USES"
R	E	Appears in "the" & "every"
O	T	In "the" & "Protect"
M	R	Fits "Strong", "Protect", "report"
A	I	Confirmed by "emails"
P	"U"	Confirmed by security

Example:

OYR → The

NRCPMAOX → SECURITY

ORSG → TEAN

NOMJHT → STRONG

USOS → DATA

PQRMX → EVERY

PNRM → USER

RGSAFN → EMAILS

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Recovered Plain text:

"The security team uses strong passwords to protect data. Every users must report suspecious emails. Security... works best when everyone follows best/ safe practices"

(c)

One Limitation:

A limitation of frequency analysis is that it may not work reliably when the cipher text is short. In a short message, the number of occurence of each letter may not represent normal english letter frequencies. This can make correct subtitutions difficult to identify.

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Question (4)

Hill Cipher:

Given Matrix:

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

(a)

Determinant and invertibility Modulo 26

$$\det(K) = (3\times5)-(3\times2)$$
$$= 15-6$$
$$= 9$$

GCD

A matrix is invertibility modulo 26 if $\gcd(\det(K), 26) = 1$

$$\gcd(9, 26) = 1$$

K is invertibility modulo 26

(b)

Encryption of "DATA"

D=3, A=0, T = 19

$$\begin{bmatrix} 3 & 19 \\ 0 & 0 \end{bmatrix}$$

First Pair DA:

$$C_1 = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}\begin{bmatrix} 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 9 \\ 6 \end{bmatrix}$$

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9 = J

6 = G

So, **DA = JG**

Second Pair: TA

$$P_2 = \begin{bmatrix} 19 \\ 0 \end{bmatrix}$$

$$C_2 = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \times \begin{bmatrix} 19 \\ 0 \end{bmatrix}$$

$$C_2 = \begin{bmatrix} 57 \\ 38 \end{bmatrix}$$

Taking modulo 26:

$$= \begin{bmatrix} 5 \\ 12 \end{bmatrix}$$

5 = F

12 = M

So, **TA = FM**

Final: JGFM

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(C)

Decryption:

Cipher text = HIAT

Determinant:

$$|K| = 9$$

Multiplicative inverse of 9 modulo 26 is 3:

$$9 \times 3 = 27 \% 26 = 1$$

Inverse Matrix:

$$K^{-1} = det(K)^{-1} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$K^{-1} = 3 \begin{bmatrix} 5 & -3 \\ -2 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 15 & -9 \\ -6 & 9 \end{bmatrix}$$

Taking Modulo 26:

$$K^{-1} = \begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix}$$

Cipher text to Numbers

HI = (7, 8)

AT = (0, 19)

$$P = K^{-1} C$$

For HI =

$$P_1 = \begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix} = \begin{bmatrix} 241 \\ 212 \end{bmatrix}$$

Taking Modulo 26 :

$$\begin{bmatrix} 7 \\ 4 \end{bmatrix}$$

7 = H
4 = E

So, **HI = HE**

For AT :

$$P_2 = \begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \begin{bmatrix} 0 \\ 19 \end{bmatrix} = \begin{bmatrix} 323 \\ 171 \end{bmatrix}$$

Modulo 26 :

$$\begin{bmatrix} 11 \\ 15 \end{bmatrix} \rightarrow \begin{matrix} 11 = L \\ 15 = P \end{matrix}$$

AT = LP

Final: **HELP**

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